RePermit: solution description

What RePermit does, the use cases it covers, the data it uses, the roles of AI and people, and how it works technically · 9 November 2026

Summary

RePermit is an application with which a consultant prepares and submits a complete application for a modification to a permitted building design in Dubai. It checks the modification in advance against Dubai law and regulations, the same rules the authority applies. It determines what kind of modification it is, which rules and disciplines are affected, which documents are required and whether they are present, complete and current. What is missing, it requests through documents and the BIM model. When new information arrives it reassesses; when rules conflict or a procedure is unclear it escalates to a person; and every step is recorded with its reasons. Once the file is complete, an internal reviewer releases it and the consultant submits it to the authority.

Decisions rest with a deterministic rules engine and with people. AI advises: it classifies, reads documents, searches the BIM model and the regulations, and drafts questions. The permit decision is taken by the competent authority in its own portal; RePermit does not replace that judgement.

The rules are derived from public Dubai sources and validated by an expert. For the demo we handle one type of permit: modifications to existing apartment buildings.

1. Business context

The process

Anyone in Dubai who wants to modify a permitted, often already completed, building needs approval from the competent authority: Dubai Municipality through the Building Permits System (BPS), or in the free zones and development areas an authority such as the Dubai Development Authority (DDA). The route depends on the authority, on whether the building has been completed, and on the kind of modification: renovation, modification, reconstruction, change of use or addition. An engineering firm (the consultant) prepares the application and is the authority's point of contact.

The problem

- Unclear what is required. The required documents differ per case: an owner NOC, structural calculations, a Civil Defence NOC, an assessment of the affected area, a planning exemption. This is spread across laws, circulars, service descriptions and the Dubai Building Code (over 1,600 passages on its own), partly only in Arabic.
- Incomplete applications and extra rounds. Missing or wrong documents lead to rejection and resubmission. Internationally this is the step where automation pays off most: in Bellevue (US) the share of applications complete at first submission rose from 22% to 67% in a test.
- A change has knock-on effects. A relocated wall looks harmless but later turns out to be a fire separation. Earlier documents and approvals then no longer hold. Without systematic impact analysis this is discovered late.
- Little traceability. Afterwards it is often impossible to see which rule, document and person led to a step.

What RePermit does about it

- Checks the modification in advance against Dubai law and regulations: Local Order 3/1999, the Civil Defence law, the Dubai Building Code and the authority's service descriptions.
- Determines the type of modification, the affected disciplines and the applicable rules, with source and article.
- Derives which documents and facts are required, and checks whether they are present, complete, of the right revision and valid.
- Requests what is missing in plain language: which document, why, and whose turn it is.
- Reassesses as soon as new information arrives, and moves the case back when earlier conclusions no longer hold.
- Escalates to a person when rules conflict, the procedure has a gap or there are risks the rules do not cover, instead of inventing a rule.
- Records every step: who or what decided, on the basis of which rules, documents and sources.
- After internal release, delivers a complete, substantiated file to submit to the authority.

Who it is for

For the consultant or permit coordinator on the applicant's side, who wants to know before submitting how the authority will assess the application, and to submit a complete one. Within the firm, an internal reviewer, for example a senior engineer, releases the file. The authority receives a complete file with substantiation per rule and document, and takes the decision itself. In August 2026 Dubai Municipality announced its own AI agent that is to issue villa permits automatically. RePermit sits on the other side of that counter: it makes sure that what comes in is complete and compliant.

2. Scope of the demo

The demo covers one type of permit: a modification to an existing, completed apartment building under the Dubai Development Authority. This lets us show the whole process, from submission to decision, with one coherent set of rules, documents and BIM models. The demo cases take place in a synthetic 12-storey apartment building.

In the demo	Outside the demo
Changes within an apartment: moving walls, merging or splitting apartments, floors, ceilings and finishes	Villas and single-family houses
Changes in common areas: lobby, corridors, escape routes and building services	New construction and new permits
Changes that affect the building: load-bearing walls, fire separations, facade	Offices, shops and other non-residential buildings
Edge cases on the outside: enclosing a terrace, a rooftop extension	Buildings under Dubai Municipality or other authorities, heritage

Two existing AI test cases concern offices (an office fit-out and converting offices to apartments); they fall outside this scope and remain available only as examples.

3. Use cases

RePermit is built on UC-01 Process a Building-Permit Design Modification (use case specification version 1.0, 18 October 2026). The primary actor is the consultant or coordinator; the authority is simulated.

Basic flow (H0)

Step	What the use case requires	How RePermit does it
H0.1	Consultant submits the modification	Description, revision and optionally the type; the IFC model is uploaded and compared with the approved model
H0.2	Retrieve project state, drawings and evidence	The file is built from the event log: every document with version and revision
H0.3	Determine rules and affected domains	The rules engine applies the rule set; AI classifies and assesses per domain, with source passages
H0.4	Determine affected documents and approvals	Per document: still valid, to be checked, or to be revalidated
H0.5	Check evidence for completeness, consistency and revision	Checklist per document type, revision and validity checks, contradictory facts
H0.6	Plan the next step	Next action and the roles allowed to perform it
H0.7	Check autonomy and authorization	May the system do this itself, or is a person needed; the role is checked
H0.8	Execute or hand over the action	Prepare the package, release it internally and submit it to the authority, by authorized roles only; in the demo the authority's decision is simulated
H0.9	Record everything	Every step in the event log; can be replayed and is explained per step in the Flow

Exceptions (E1–E6)

Flow	Situation	How RePermit responds
E1	Evidence missing or insufficient	The step is held; the missing document is requested, with an upload button at the item itself
E2	New design revision	Earlier evidence is marked as belonging to an old revision; impact is determined again
E3	Conflicting rules	The system does not choose; escalation to the internal reviewer, who asks the authority for clarification if needed
E4	Role may not perform the action	Action refused and recorded; the case does not change
E5	Late material information	The case returns to the start of the stage; earlier conclusions lapse
E6	Gap in procedure or authority	Escalation instead of an improvised procedure

Supporting functions

- BIM: reading IFC models, comparing them with the approved model, deriving facts (load-bearing, fire separation, facade) and letting the AI query them through tools.
- Documents: uploading, storing and opening PDFs; a checklist per document type; the AI reads, the code decides whether a document is complete, and a person confirms the facts.
- Review rounds: add information and have it reassessed, at most four rounds; after that a person takes over.
- Cost control: hard limits per case ($1) and per day ($5); tokens and costs recorded per AI call.
- Settings: model and effort per AI component, number of BIM tool calls, retrieval settings and budgets, with a change history.
- Knowledge: the rules, sources and the searchable knowledge index can be viewed in the app.
- Demo: 9 fixed test cases for the flows and 8 AI cases in the apartment building, from replacing floors and a wall between two apartments to a rooftop extension.

4. Data and processing

RePermit uses six kinds of data. In the demo all cases, documents and models are synthetic.

Data	Origin	Processing	Storage
Rule set	Derived by hand from public sources: Local Order 3/1999, Law 4/2025 (Civil Defence), Law 6/2019, the Dubai Building Code, DDA service descriptions and circulars	20 rules, 7 known gaps, 16 document types with checklists, 5 modification types and 6 domains; every rule with source and article; interpretations labelled	YAML in the code, versioned (for example dubai-public-2026.4); each case keeps its version, old versions are retained
Knowledge index	Full texts of 15 public sources plus 14 own summaries (knowledge cards)	Downloaded, split per article with authority and validity, turned into embeddings by a local model; hybrid search, at most 5 articles and 3 cards per review (see section 6)	Vector database (PostgreSQL with pgvector) plus a keyword index, within our own environment
BIM models	IFC files from the applicant; in the demo a synthetic 12-storey apartment building	Read with IfcOpenShell; per element the class, storey, properties, quantities and material; compared with the approved model by GlobalId; facts derived	In the event log with the evidence item
Documents	PDFs from the applicant (synthetic in the demo)	Text extracted; each checklist item assessed by the AI or the internal reviewer; facts only count after confirmation	Text in the event log, the file itself by checksum (SHA-256)
Answers and notes	Entered by the applicant	Only the opening question (competent authority) counts directly; other text is context and proves nothing	Event log

Data	Origin	Processing	Storage
Event log	Every step in every case	Each event is appended, never changed; the status follows from the sequence; every step can be replayed	PostgreSQL, with daily backups

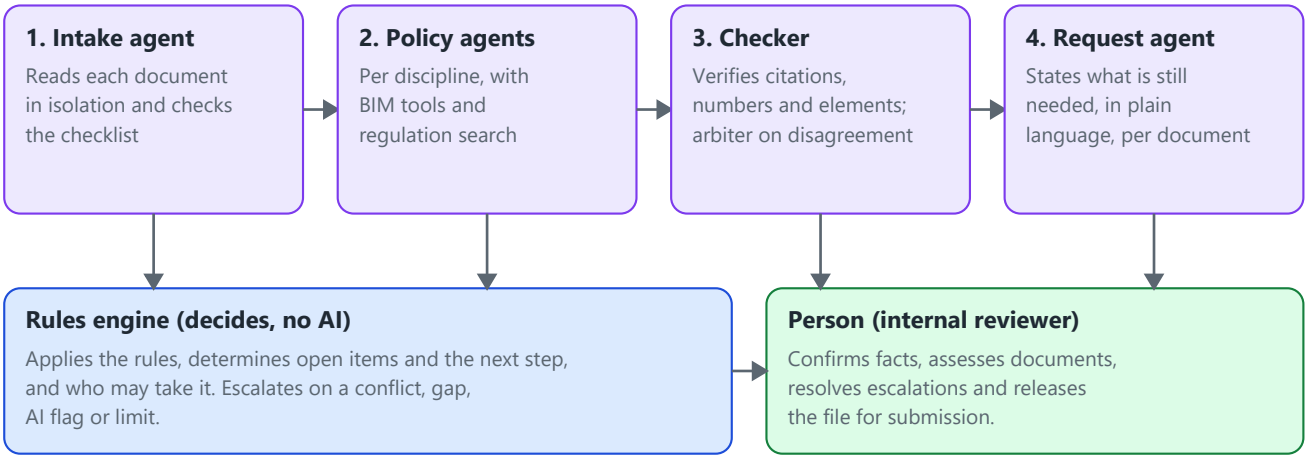
What goes to the AI

For each review RePermit sends to the AI model (TBD, sandbox): the description of the modification, the established facts, the evidence metadata, the text of documents not yet read, the model comparison, the source passages and the answers from the BIM tools. The full IFC file and documents already assessed are not sent. Each call records the model, the tokens and the estimated cost. Text from documents and models is treated as data, not as instructions.

5. Agents and the role of people

Agent architecture

RePermit works with a fixed sequence of specialised agents around one shared case file. This design follows from research into existing systems (CivCheck in Seattle, CORENET X in Singapore, recent academic studies). The agents do not talk freely to each other; the code sets the order, and the rules engine and people decide.



All steps read from and write to one shared case file (the event log). Purple = AI (advises), blue = rules, green = person.

Agent	Does	Does not	Model
Intake agent	Reads each new document separately, checks type, case and revision, works through the checklist, extracts facts with a quote	No tools, no decision on completeness	TBD (sandbox)
Policy agents	Per discipline (structure, fire, planning, property): assesses with BIM tools and regulation search, citing sources	Apply rules or deviate from the rule set	TBD (sandbox)
Checker	Verifies that citations, numbers and BIM elements are right; arbiter only when AI and rules disagree	Draw conclusions of its own	TBD (sandbox)

Agent	Does	Does not	Model
Request agent	Turns open items into a clear request: which document, why, whose turn	Treat anything as proven	TBD (sandbox)

What people do

Role	Responsibility
Consultant / coordinator	Submits the modification, provides model and documents, answers questions, prepares the package and submits it
Internal reviewer	Senior engineer or quality controller at the consultant. Confirms or rejects facts the AI read from documents, assesses documents directly, resolves escalations and releases the file for submission
Competent authority	DDA or Dubai Municipality: assesses the submitted application in its own portal and takes the decision. Simulated in the demo
Observer	Follows the case, without actions
Rule set expert	Validates rules and interpretations, approves proposals for new rules
Administrator	Sets models, budgets and retrieval settings within the limits of the server

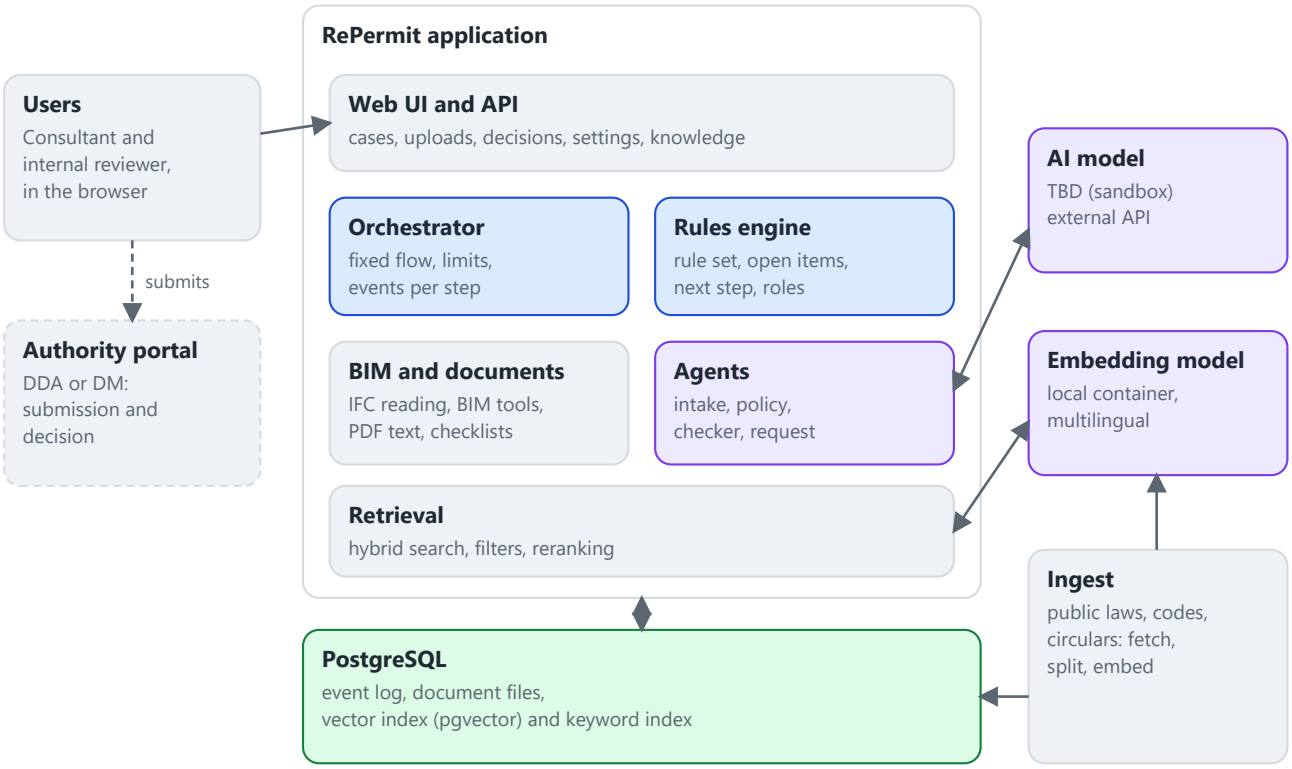
Principles: the AI advises, the rules decide; the AI may speed up a 'yes', every 'no' goes through a person; every step is visible in the Flow with who decided and on what basis. The permit decision stays with the authority; RePermit makes sure it receives a complete, substantiated file.

6. Technical design

This section describes how a case moves through RePermit technically, how the AI steps are set up and how the legal and regulatory texts are made searchable through embeddings.

Architecture

RePermit runs in our own environment: the application, a local embedding model and PostgreSQL. Only the AI model is an external service. Users work in the browser; the application prepares the file, and submission and the decision take place in the authority's portal.



Grey = external or supporting, blue = deterministic, purple = AI, green = data. Dashed = outside RePermit.

Flow of a case

Step	Who	What happens technically
1. Submission	Applicant	The modification and IFC model arrive. The app reads the model, compares it with the approved model and derives facts such as load-bearing, fire separation and facade
2. Retrieve knowledge	Retrieval	A search query is built from the case. The vector database returns the best-matching articles and knowledge cards, filtered by authority and validity
3. AI steps	Agents	Intake reads the documents, the policy agents assess per discipline with BIM tools and the retrieved articles, the checker verifies their outcome
4. Rules check	Rules engine	Applies the rule set and determines open items, conflicts and gaps, the next step and who may take it
5. Requests and release	People	The consultant provides what is missing. The internal reviewer confirms facts, resolves escalations and releases the file; then it is submitted to the authority. New information restarts the flow from step 2

Every step is recorded as an event in the case file. Before every AI call the app checks the limits for rounds and budget.

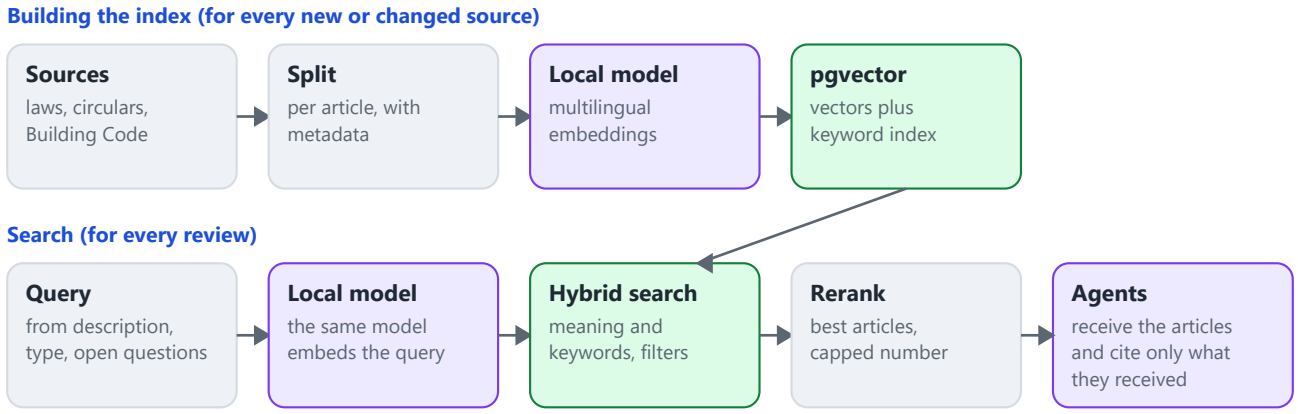
Agents

Agent	Receives	Returns	May use
Intake	New documents with their checklist	Per document: right type, belongs to the case, checklist items, facts with a quote	Nothing: no tools
Policy (per discipline)	Case, established facts, model comparison, retrieved articles	Findings per domain, questions, citations	BIM tools: overview, find elements, inspect an element, what is nearby
Checker	Outcome of the policy agents and the evidence	Agreement or deviations with reasons	The same BIM tools, read only
Request	Open items from the rules engine	Request to the applicant in plain language	Nothing

All agents return a fixed response schema. Each agent has its own model, effort and limit, set on the Settings page, and records its tokens and cost. Text from documents and models is data, never instructions.

Embedding of legal and regulatory texts

The legal and regulatory texts are turned into vectors by a local embedding model and stored in a vector database. A local model keeps the source texts within our own environment, which is required because they may only be used internally.



Building the index

- Fetch: the sources on the source list (laws, circulars, Dubai Building Code, service descriptions) are downloaded; date, version and terms of use are recorded per source.
- Split: per article or clause, with the chapter heading added as context. Each chunk gets metadata: source, article number, authority (DDA, DM or all), in force or superseded, language and source version.
- Embed: a multilingual open model (for example from the BGE or E5 family) runs in its own container and turns each chunk into a vector. English and Arabic texts share the same vector space.
- Store: in PostgreSQL with the pgvector extension, next to a keyword index on the same chunks, so no separate database is added.
- Update: when a source changes, only that source's chunks are read and embedded again. Superseded articles are kept but no longer returned.

Search

- Hybrid: search by meaning (vectors) and by keywords, with the results merged. Legal texts contain exact terms and references, such as *Art. 17(4)* or *EI60*, that only keyword search finds reliably.
- Filters first: only chunks that apply to the competent authority, are still in force and come from an enabled source.
- Rerank: the best candidates are reordered and the number is capped at what is set on the Settings page (5 articles and 3 knowledge cards by default).
- Check: an agent may only cite chunks it received; the app removes any other citation. The retrieved chunks are shown per review in the Flow and can be browsed on the Knowledge page.
- Measure: a fixed set of test questions with the right articles. Keyword-only search reached a precision of about 75%; the vector index must demonstrably do better.

7. Design principles

- RePermit does not decide on a permit; the competent authority does. RePermit delivers a complete, substantiated file.
- The AI advises. What counts is established by the rules engine or the internal reviewer.
- The rule set translates law and regulation into conditions. Every rule and interpretation is validated by an expert and versioned; a new version applies only to new cases.
- The full source texts stay within our own environment; hence a local embedding model.
- Documents need a text layer; scanned documents are outside the first version.
- In the first version, submission takes place in the authority's portal; a direct integration follows once the authority offers one.
- See also the reports *RePermit: what it takes to mature the product* and *Agent architectures for building permits*.